TEXORA · LECTURE SHEET

Control of Yarn Realization

Yarn realisation is the mill's master material account: of all the cotton issued, how much left as yarn? This sheet defines realisation, builds the department waste budget, and shows how the difference between expected and actual output — the invisible loss — is isolated and hunted down.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 11 · SPECIALIZATION — YARN MANUFACTURING

1. Definitions

$$\text{Yarn realisation (\%)} = \text{yarn produced (kg)} / \text{raw material issued (kg)} \times 100$$

The complement is total waste. Waste divides into **visible waste** — everything collected and weighed: blowroom droppings, card waste, comber noil, fly, sweepings, hard waste from winding — and **invisible loss**, found only by difference: moisture change, dust exhausted, spillage, weighing errors, pilferage. Some visible waste is reusable in the same mixing (soft waste such as roving ends and pneumafil waste, within limits) and some is saleable (noil, droppings); realisation accounting must state which convention it uses. The convention here: realisation counts only yarn out against material in; reusable waste returned to the process is credited back to input.

2. The expected realisation — the waste budget

Each department's waste norm multiplies into an expected realisation (the same arithmetic as the spin plan, YM-3207 Topic 8). For a carded-yarn month the budget is:

STAGE	WASTE NORM	RETENTION	CUMULATIVE RETENTION
Blowroom	5.8%	0.942	0.942
Card	4.2%	0.958	0.902
Drawframe + simplex	0.8%	0.992	0.895
Ring frame	1.2%	0.988	0.884
Winding	0.5%	0.995	0.880

WORKED EXAMPLE — ISOLATING THE INVISIBLE LOSS

Month's issue of mixing = 52,400 kg. Expected output from the budget = 52,400 × 0.880 = **46,115 kg**.

Actual yarn produced and weighed = **45,580 kg** → actual realisation = 45,580/52,400 × 100 = **87.0%** against a standard of 88%.

Shortfall = 46,115 − 45,580 = **535 kg**, i.e. invisible loss = 535/52,400 × 100 = **1.02%** of input.

Interpretation: visible wastes ran at their norms (the budget matched collection records), so the missing 535 kg left unweighed — moisture loss in a dry month, dust extraction, spillage at transport, or scale error. 1.02% sits at the edge of the customary ≲ 1% tolerance: the month is flagged, and the investigation order is scales first (cheapest to check), then moisture (issue and yarn weighed at different regains — see YM-3207 Topic 6), then dust and spillage.

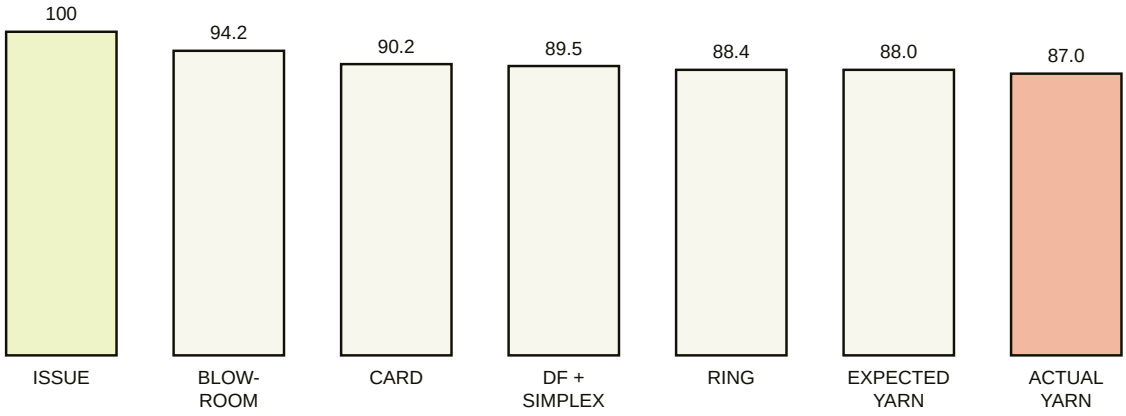


Figure 1 — Realisation waterfall for the worked example: cumulative retention after each department (per 100 kg issued). The final step from expected 88.0 to actual 87.0 is the invisible loss.

3. Department waste control

- **Weigh everything, both ways:** waste is weighed where it arises and again at the waste store; discrepancies localise pilferage and scale faults.
- **Reusable waste discipline:** soft waste returned to the mixing is capped (commonly a few percent of the mix) because over-return degrades quality: it has already been opened and worked once.
- **Hard waste** (winding and ring hard waste, contaminated sweepings) must never re-enter the mixing; it is sold. Its weight trend is also a process signal — rising hard waste means rising breaks somewhere.
- **Moisture honesty:** realisation compares masses weighed weeks apart in different departments; in humid months yarn can gain regain and flatter realisation. Serious mills correct both sides to standard regain (YM-3207 Topic 6 method) before judging a small deviation.

4. Acting on realisation

Realisation is a monthly integrator of every department's discipline, so it is reviewed monthly with the department waste table beside it. The diagnostic logic: actual below expected with one department's visible waste above norm → act in that department (its own topic sheet). All visible wastes at norm but actual low → invisible loss hunt, in the order: scales, moisture, dust extraction, spillage, records. Realisation above expectation is also investigated — it usually means waste is being under-weighed or yarn over-weighed, and the customer will find the error if the mill does not.

Glossary

Yarn realisation — yarn output as % of raw material issued. **Visible waste** — waste collected and weighed at each process. **Invisible loss** — the unweighed difference: moisture, dust, spillage, errors. **Soft waste** — clean, reusable process waste (roving ends, pneumafil waste). **Hard waste** — twisted, contaminated waste that cannot be respun in the same mixing. **Waste budget** — the expected realisation built by multiplying department retentions. **Noil realisation note** — combed mills quote realisation with noil credited separately, since noil is a saleable product, not a loss.

Key points

• Realisation = yarn out / material in × 100; its complement splits into visible waste and invisible loss.
• Budget example: retentions multiply to 0.880 → expected 46,115 kg from 52,400 kg issued.
• Actual 45,580 kg = 87.0%; the 535 kg gap is invisible loss at 1.02% of input — flagged at the ≲1% tolerance edge.
• Hunt order for invisible loss: scales, moisture/regain differences, dust, spillage, records.
• Reusable waste is capped in the mixing; hard waste never returns; realisation above standard is also an error signal.

Practice

1. A mill issues 30,000 kg and produces 26,700 kg of yarn, with all visible wastes exactly at budget (expected realisation 90.5%). Calculate actual realisation and the invisible loss in kg and %.
2. Blowroom waste rises 1.5 points above norm while other departments hold theirs. By how much does expected realisation fall, approximately, if the budget was 88.0%?
3. Why can a humid month show realisation above standard with no change in process performance?

Answer key

1. Actual realisation = 26,700/30,000 × 100 = **89.0%**. Expected yarn = 30,000 × 0.905 = 27,150 kg. Invisible loss = 27,150 − 26,700 = **450 kg = 1.5%** of input — above tolerance; investigate scales and moisture first.
2. The blowroom retention falls from (say) 0.942 to 0.927, a factor of 0.927/0.942 = 0.984. Expected realisation ≈ 88.0 × 0.984 = **86.6%** — about 1.4 points lost, showing how an early-stage waste multiplies through the whole account.
3. Yarn weighed at despatch in humid air carries more regain than the cotton had when issued; the extra water is weighed as yarn. Correcting both weighings to standard regain removes the false gain.

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